

Capstone Project Proposal

Course: BA-64099

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Project Title: Heart Disease Risk Prediction Using Clinical and Lifestyle Indicators

1. Data Source:

The dataset selected for this project is the *Heart Failure Prediction Dataset* available on Kaggle: <https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>

This dataset contains anonymized patient records with a variety of clinical and behavioral attributes, including age, blood pressure, cholesterol levels, chest pain type, fasting blood sugar, ECG results, maximum heart rate, and exercise-induced angina. The binary target variable indicates whether a patient has heart disease (1) or not (0).

2. Project Objective:

The objective of this project is to develop a binary classification model that can predict an individual's likelihood of having heart disease based on their medical and lifestyle data. By leveraging predictive analytics, the goal is to support early identification of high-risk patients, thereby facilitating timely medical intervention and preventative care.

3. Problem Statement:

Cardiovascular disease remains one of the leading causes of death globally. Early detection is critical for improving patient outcomes. This project seeks to build a supervised learning model to classify individuals as either “at risk” or “not at risk” for heart disease, based on readily available clinical indicators. The output will be a probabilistic or binary prediction to assist in early screening processes.

4. Methodology Overview:

Data Preparation

- Perform exploratory data analysis (EDA) to understand feature distributions and relationships
- Encode categorical variables (e.g., chest pain type)
- Detect and handle outliers
- Normalize numerical variables as needed
- Conduct feature selection based on domain relevance and correlation analysis

Modeling Techniques

- Logistic Regression for baseline performance and interpretability
- Decision Tree Classifier for simple decision rules
- Random Forest Classifier to improve predictive performance and handle feature interactions

Model Evaluation

- Accuracy
- Precision and Recall
- F1-Score
- ROC-AUC Curve
- Confusion Matrix

6. Expected Outcomes:

- A predictive model capable of identifying individuals at risk of heart disease with high accuracy
- Insights into the most influential features contributing to heart disease risk
- A demonstration of how data-driven methods can support early intervention and improve public health decision-making